



10116 - The shape of Little Red Dots: a MIRI-MRS pilot

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	GN-9771	MIRI Medium Resolution Spectroscopy	(1) GN-9771

ABSTRACT

JWST has uncovered a population of intriguing accreting black holes (BHs) called ‘Little Red Dots’ (LRDs), characterized by a v-shaped continuum, broad-line regions (BLRs) in hydrogen and helium, and a ‘soft’ Balmer break produced by absorption in the same hydrogen and helium BLRs,

indicative of extremely dense gas in the immediate vicinity of the accreting BH. This peculiar novel class of active BHs likely represents an important phase of early supermassive BH growth, but it also poses new challenges: the nature of the dense gas cocoon around the BH is not understood. Several works have suggested that scattering processes in this dense medium could significantly inflate the apparent widths of BLRs. This would have important implications on our current interpretation of high- z BH masses, as they would be significantly overestimated by up to two orders of magnitude. A decisive test of the suggested scenarios is to obtain BLR spectra from transitions to different levels of the same atom, because this is where the predictions vary. We propose a pilot study with MIRI-MRS to map the BLR with Paschen lines for a luminous LRD, to compare to the available, high-S/N Balmer lines. The results from this study will form an important contribution to the ongoing debate regarding the nature of LRDs, and pave the way for future MIRI-MRS studies of these enigmatic objects.

OBSERVING DESCRIPTION

This program consists of MIRI-MRS observations of a compact broad-line AGN at $z=5.5$, which has ancillary NIRCAM imaging and WFSS data, as well as NIRSpec-IFS R100 and R2700 data. The goal is to obtain Pa-alpha in the SHORT grating in Channel 3, to compare the line flux and shape to the existing H-alpha and H-beta data. This requires MIRI observations in the medium-resolution mode. The target is located in a well-studied extragalactic deep field containing stars with good astrometry, and therefore no target acquisition is required. Small inaccuracies in acquisition are unproblematic because of the large 5.2"x6.2" FoV of MIRI Channel 3. We use the SLOWR1 readout mode with four 4-point dithers (alternating negative, positive), with 7 integrations per exposure and 20 groups per integration, such that individual integrations are <500s long, to maximise exposure time while limiting the impact of cosmic rays. The total exposure time including overheads is modest with 19.5 h (14.9 h on source). No background exposures are required since each observations contains a sufficient number of spaxels free from emission from the central source or neighbours, and we will use spaxels within the IFS FoV free from line emission to perform a background subtraction. We make use of simultaneous MIRI imaging at no extra cost to obtain accurate astrometry.

Proposal 10116 - Targets - The shape of Little Red Dots: a MIRI-MRS pilot

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GN-9771	RA: 12 37 7.4400 (189.2810000d) Dec: +62 14 50.31 (62.24731d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO				

Proposal 10116 - Observation 1 - The shape of Little Red Dots: a MIRI-MRS pilot

Observation	Proposal 10116, Observation 1: GN-9771 Diagnostic Status: Warning Observing Template: MIRI Medium Resolution Spectroscopy					Fri Mar 13 18:05:53 GMT 2026
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.					
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous	
	(1)	GN-9771	RA: 12 37 7.4400 (189.2810000d) Dec: +62 14 50.31 (62.24731d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO			
Acquisition	#	Target				
	1	NONE				
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction	
		Channel 2	YES	FULL	Allow Auto Reorder	
Dithers	#	Dither Type	Optimized For	Direction		
	1	4-Point	POINT SOURCE	POSITIVE		
	2	4-Point	POINT SOURCE	NEGATIVE		
	3	4-Point	POINT SOURCE	POSITIVE		
	4	4-Point	POINT SOURCE	NEGATIVE		

Proposal 10116 - Observation 1 - The shape of Little Red Dots: a MIRI-MRS pilot

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exposure	Exposures/Dithers	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	20	7	1	Dither 1	4	28	13951.713	
	1	SHORT(A)	MRSLONG		SLOWR1	20	7	1	Dither 1	4	28	13951.713	276268
	1	SHORT(A)	MRSSHORT		SLOWR1	20	7	1	Dither 1	4	28	13951.713	276268
	2		IMAGER	F560W	SLOWR1	20	7	1	Dither 2	4	28	13951.713	
	2	SHORT(A)	MRSLONG		SLOWR1	20	7	1	Dither 2	4	28	13951.713	276268
	2	SHORT(A)	MRSSHORT		SLOWR1	20	7	1	Dither 2	4	28	13951.713	276268
	3		IMAGER	F560W	SLOWR1	20	7	1	Dither 3	4	28	13951.713	
	3	SHORT(A)	MRSLONG		SLOWR1	20	7	1	Dither 3	4	28	13951.713	276268
	3	SHORT(A)	MRSSHORT		SLOWR1	20	7	1	Dither 3	4	28	13951.713	276268
	4		IMAGER	F560W	SLOWR1	20	7	1	Dither 4	4	28	13951.713	
	4	SHORT(A)	MRSLONG		SLOWR1	20	7	1	Dither 4	4	28	13951.713	276268
	4	SHORT(A)	MRSSHORT		SLOWR1	20	7	1	Dither 4	4	28	13951.713	276268